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Roll No

FT-7001-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Fire Fighting Installation

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Discuss function of automatic sprinkler system. Explain the various performance characteristics of sprinkles.
b) What are the criteria of spacing between sprinklers head in different hazards category.
2. What are the various temperature ratings available for sprinklers heads based on standardized test? What are the four categories of sprinkler systems?
3. a) Explain the physical properties that allow water to be an effective extinguishing agent.
b) What are the three important methods widely used to determine the manual water application rate needed to extinguish a fire?
4. a) Write the function of water monitor in fire fighting system.
b) With a neat sketch explain the function of a Foam Pourer system. Explain the limitations on the use of low expansion foams.

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5.
 - a) Define protection areas and maximum spacing for different degree of hazard in sprinkler system as per NFPA-13. Explain typical distribution pattern from a standard spray sprinkler.
 - b) What are the uses and limitations of medium and high expansion foaming agents SYNDET? Which type of foam concentrate is used for toxic vapor suppression.
6.
 - a) Explain the performance requirement Carbon dioxide fire extinguisher as per IS:2878 and also discuss on maintenance and check list of CO₂ fire extinguisher.
 - b) Explain the application of automatic fire control systems used for flammable liquid storage tanks.
7. Explain the foam based fire fighting systems base on :
 - i) Type of fuel
 - ii) Burning depth of fuel
 - iii) Size of fire
 - iv) Application rate and discharge time
8. Write short notes on the followings.
 - a) Escape route during fire
 - b) ISU and ISO methods
 - c) Deluge system
 - d) Usefulness of Earthing and Bonding
